

## SUPPLEMENT TO “NEURAL BELLMAN OPERATORS”

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This supplement provides the operator proofs, verification conditions, transition construction, additional numerical results, and replication instructions for the sixth revision. It distinguishes continuous-time economic objects from finite transition approximations and distinguishes trained neural policies from analytical and finite-state regression tests. The main paper and this supplement use the same policy-evaluation, boundary, and temporal-self conventions.

## S.1. OPERATOR PROOFS AND QUANTITATIVE APPROXIMATION

S.1.1. *Exact Policy Improvement*

We use the assumptions of the main paper’s exact separated-operator proposition. In particular,  $I(p)(t, s)$  is a selector maximizing  $\mathcal{H}^a[E(p)](t, s)$ , not a stationary point of a parameterized actor objective. Continuity of  $E$  is in the norm controlling the value, time derivative, first spatial derivatives, second spatial derivatives, and the stated boundary traces. The admissible compact policy class is closed under  $I$ . These are assumptions on an exact boundary-value problem, not universal properties of neural policy classes.

PROOF OF EXACT SEPARATED-OPERATOR CONSISTENCY: Write  $V_n = E(p_n)$  and  $p_{n+1} = I(p_n)$ . The evaluation equation gives  $V_{n,t} + \mathcal{H}^{p_n}[V_n] = 0$ . Pointwise maximization implies

$$V_{n,t} + \mathcal{H}^{p_{n+1}}[V_n] \geq 0.$$

The assumed comparison and verification inequalities, with the same boundary operator for both policies, imply  $V_{n+1} \geq V_n$ . Precompactness of the evaluation image implies boundedness and hence a pointwise limit  $\bar{V}$ . Every subsequential limit in the declared strong topology has this same value component. Its derivative components are the derivatives of that value, because uniform convergence of the functions and their derivatives preserves differentiation. Consequently all strong accumulation points coincide and  $V_n \rightarrow \bar{V}$  strongly.

Let  $p_{n_j} \rightarrow p$  be a policy subsequence, which exists by compactness. Continuity of evaluation gives  $E(p) = \bar{V}$ . Continuity of improvement at  $p$  and continuity of evaluation give

$$E(I(p)) = \lim_j E(I(p_{n_j})) = \lim_j V_{n_j+1} = \bar{V}.$$

Evaluating  $I(p)$  at this common value yields

$$0 = \bar{V}_t + \mathcal{H}^{I(p)}[\bar{V}] = \bar{V}_t + \sup_a \mathcal{H}^a[\bar{V}].$$

The boundary traces also pass to the limit. Comparison identifies  $\bar{V}$  with the optimal value, and verification identifies  $p$  as optimal because  $E(p) = \bar{V}$ . This argument does

not require  $p = I(p)$  or convergence of the entire policy sequence. In the presence of tied maximizing actions, different optimal policies may share the same limiting value. *Q.E.D.*

The proof is an application of exact policy-improvement reasoning; it is not a claim of priority over [Jacka and Mijatović \(2015\)](#). The assumption of a compact strongly regular evaluation image can fail at corners or under degenerate diffusions. The stopped application is therefore evaluated by its specified positive transition approximation rather than asserted to satisfy every classical hypothesis globally.

### S.1.2. Finite-Horizon Value-Loss Bound

For clarity, put  $\Gamma_{n,n} = 1$  and  $\Gamma_{n,j} = \prod_{\ell=n}^{j-1} \gamma_\ell$ . Both  $\mathcal{T}_n^a$  and its policy version are monotone and  $\gamma_n$ -Lipschitz in the sup norm. The implemented operator is  $\tilde{\mathcal{T}}_n^a$ . The three nonnegative error bounds at  $\hat{v}_{n+1}$  are

$$\begin{aligned}\epsilon_n &\geq \left\| \hat{v}_n - \tilde{\mathcal{T}}_n^{\hat{p}_n} \hat{v}_{n+1} \right\|_\infty, \\ \delta_n &\geq \left\| \tilde{\mathcal{T}}_n \hat{v}_{n+1} - \tilde{\mathcal{T}}_n^{\hat{p}_n} \hat{v}_{n+1} \right\|_\infty, \\ \kappa_n &\geq \sup_a \left\| (\tilde{\mathcal{T}}_n^a - \mathcal{T}_n^a) \hat{v}_{n+1} \right\|_\infty.\end{aligned}$$

Here  $\tilde{\mathcal{T}}_n = \sup_a \tilde{\mathcal{T}}_n^a$ , and  $\|\hat{v}_N - G\|_\infty \leq \eta$ .

PROOF OF THE MAIN VALUE-LOSS THEOREM: Let  $D_n = \left\| \hat{v}_n - J_n^{\hat{p}} \right\|_\infty$ . Inserting the implemented and target fixed-policy backups gives

$$D_n \leq \epsilon_n + \kappa_n + \gamma_n D_{n+1}.$$

Let  $R_n = \|\hat{v}_n - V_n^*\|_\infty$ . The supremum inequality  $|\sup_a f_a - \sup_a g_a| \leq \sup_a |f_a - g_a|$  and the nonnegative improvement gap give

$$R_n \leq \epsilon_n + \delta_n + \kappa_n + \gamma_n R_{n+1}.$$

Backward iteration, with  $D_N, R_N \leq \eta$ , proves

$$\begin{aligned}D_n &\leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \kappa_j) + \Gamma_{n,N} \eta, \\ R_n &\leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \delta_j + \kappa_j) + \Gamma_{n,N} \eta.\end{aligned}$$

Feasibility implies  $V_n^* \geq J_n^{\hat{p}}$ . Combining the two inequalities proves the stated upper bound with coefficients  $2\epsilon_j + \delta_j + 2\kappa_j$  and terminal coefficient  $2\eta$ . *Q.E.D.*

The theorem is finite-horizon. It permits a recursive operator with  $\gamma_n > 1$  but then records amplification explicitly through  $\Gamma_{n,j}$ . It does not imply contraction for an arbitrary Epstein–Zin aggregator. Boundary killing can be included in the transition kernel by treating the boundary payoff as a known reward component; only continuation mass contributes to  $\gamma_n$ .

### S.1.3. Differential Residual and Boundary Error

Let  $v$  be a  $C^{1,2}$  candidate for the additive, stopped, discounted problem. Define  $R^a = v_t + \ell^a + \mathcal{L}^a v - \rho v$ . Suppose  $|R^\beta| \leq \epsilon(t)$ ,  $\sup_a R^a - R^\beta \leq \delta(t)$ , and the payoff discrepancy at terminal time or exit is at most  $\eta$ . Assume the localized Itô formula passes to expectations by uniform integrability.

For every admissible  $p$ , stopped Itô integration gives

$$J^p(t, s) - v(t, s) = \mathbb{E}_{t,s}^p \left[ \int_t^\tau e^{-\rho(r-t)} R^{p_r}(r, S_r) dr + e^{-\rho(\tau-t)} (G - v)(\tau, S_\tau) \right].$$

The upper bound for an arbitrary policy is  $\eta + \int_t^T e^{-\rho(r-t)} (\epsilon(r) + \delta(r)) dr$ . The absolute discrepancy for  $\hat{p}$  is at most  $\eta + \int_t^T e^{-\rho(r-t)} \epsilon(r) dr$ . Taking a supremum over  $p$  and adding proves the differential corollary. The argument is conditional on uniform residual and boundary bounds. Replacing these bounds by sampled root mean squared losses is not a valid proof step.

### S.1.4. Feasible Candidates, Convex Gaps, and Reference Bounds

If  $Q(s, a)$  is  $L_a$ -Lipschitz in  $a$  and a feasible set  $\mathcal{C}(s)$  is an  $h_a$ -net of the admissible actions, then a maximizer over  $\mathcal{C}(s)$  is within  $L_a h_a$  of the full supremum. Adding a neural proposal to  $\mathcal{C}(s)$  cannot reduce its achieved backup value. Neither statement bounds the evaluation error of the critic that generated  $Q$ .

For differentiable convex  $F$  on a compact convex set  $A$ , its tangent inequality implies

$$F(a) - \inf_{b \in A} F(b) \leq \nabla F(a)^\top a - \min_{b \in A} \nabla F(a)^\top b =: g_F(a).$$

For  $A = \{a \geq 0 : \sum_i a_i \leq B\}$ , the minimum is  $B \min(0, \min_i \partial_i F(a))$ . The same formula applies to a scenario tree with separate feasible actions at each information node: sum the node-specific linear minimization gaps, using the gradient of the complete probability-weighted objective. This gradient must include all descendant costs.

If a feasible reference has objective  $C_R$  and gap  $G_R$ , then  $C_R - G_R \leq C^* \leq C_R$ . Any other feasible policy, evaluated on the same tree, therefore satisfies

$$C_\pi - C_R \leq C_\pi - C^* \leq C_\pi - C_R + G_R.$$

The program evaluates both objectives by all 64 terminal branches. It does not insert the neural critic as a substitute for policy cost. The reported interval is based on ordinary double-precision evaluation of these inequalities; it is not an interval-arithmetic enclosure of rounding error.

A useful conditional policy bound also follows. If the optimal continuation backup  $Q^*(s, \cdot)$  is  $\mu$ -strongly concave and the approximate continuation is within  $e$  of the optimal one, an action whose approximate improvement gap is at most  $\delta$  obeys

$$\|\hat{a} - a^*\|^2 \leq \frac{2}{\mu} (\delta + 2\gamma e).$$

Indeed the value of each action changes by at most  $\gamma e$ , and strong concavity bounds the resulting true suboptimality below by  $\mu \|\hat{a} - a^*\|^2 / 2$ . This is not applied to a nonconcave NDU maximand. In the convex resource problem the immediate quadratic cost

supplies strong convexity  $0.2/d$  for minimization. Its deterioration with dimension is included in the conditioning discussion rather than hidden in a dimension-free speed claim.

## S.2. COUNTEREXAMPLES, TRACE ESTIMATION, AND DIFFERENTIATION

### S.2.1. *Why a Joint Penalty Does Not Define the Economic Solution*

Consider  $-V' + V = 0$  on  $[0, 1]$ ,  $V(1) = 1$ . Its solution is  $V^*(t) = e^{t-1}$ . The historical composite objective was of the form

$$L(V) = \int_0^1 \{V(t) + [-V'(t) + V(t)]^2\} dt.$$

The feasible perturbation  $V_\varepsilon(t) = V^*(t) - \varepsilon(1 - t)$  preserves the terminal condition but has residual  $-\varepsilon(2 - t)$ . Consequently

$$L(V_\varepsilon) - L(V^*) = -\frac{\varepsilon}{2} + \frac{7\varepsilon^2}{3} < 0 \quad \text{for } 0 < \varepsilon < \frac{3}{14}.$$

Thus the exact economic solution is not even a local minimizer of this penalty. Separate parameter blocks, objectives, and verification conditions are substantive changes, not a change of notation for that joint loss.

### S.2.2. *A Stable Actor Stationary Point Need Not Be Optimal*

On  $[-2, 2]$ , let  $H(a) = -(a^2 - 1)^2 + 0.2a$ . Its stationary equation is  $4a^3 - 4a - 0.2 = 0$ . There are three real roots. At the negative local maximizer, approximately  $-0.9739943532$ ,  $H''(a) = 4 - 12a^2 < 0$ , so gradient ascent has a locally asymptotically stable equilibrium. The positive maximizer is approximately  $1.0241203002$  and has higher payoff by approximately  $0.3998745872$ . Both are isolated. The example embeds into a finite-horizon control problem with flow  $H(a)$  and exact evaluation  $(T - t)H(a)$ .

Consequently, even exact fast evaluation, diminishing separated step sizes, bounded iterates, and zero noise do not turn isolated actor stationary points into policy-improvement fixed points. Under a suitable stochastic-approximation theorem the justified conclusion concerns the limiting invariant or stationary set. Global optimality additionally requires an argument excluding suboptimal attractors. Convex improvement in the resource example supplies such an argument for each surrogate action problem; it does not automatically eliminate critic approximation error. Constant-step Adam has no convergence theorem in this paper, and the executed neural runs use the explicitly recorded L-BFGS and constrained least-squares routines instead.

### S.2.3. *Small Strong Residuals Can Miss a Viscosity Solution*

For the exit-time problem  $dX_t = a_t dt$ ,  $|a_t| \leq 1$ , with flow  $-1$  and zero payoff on  $\{-1, 1\}$ , the value is  $V(x) = |x| - 1$ . The viscosity equation is  $1 - |V'| = 0$ . The smooth functions

$$W_\varepsilon(x) = \sqrt{1 + \varepsilon^2} - \sqrt{x^2 + \varepsilon^2}$$

satisfy the boundary data, while their squared strong residual under the uniform distribution tends to zero:

$$\frac{1}{2} \int_{-1}^1 \left(1 - \frac{|x|}{\sqrt{x^2 + \varepsilon^2}}\right)^2 dx \sim \left(2 - \frac{\pi}{2}\right) \varepsilon.$$

Nevertheless  $W_\varepsilon \rightarrow 1 - |x|$ , which has the wrong sign and is not the desired viscosity solution. The asymptotic expression follows by  $x = \varepsilon y$  and integration of the resulting integrable function on  $[0, \infty)$ . The code also evaluates the finite- $\varepsilon$  integral by independent adaptive quadrature.

Monotonicity, stability, and consistency with an appropriate comparison principle are central to viscosity approximation; see [Barles and Souganidis \(1991\)](#). Smooth network approximation of a function and its derivatives, as in [Hornik, Stinchcombe, and White \(1990\)](#), concerns a smooth target and does not approximate a derivative at a kink or replace a viscosity-selection argument. Historical kink and residual arrays remain available as archival material, not as evidence that an unregularized strong-residual loss selects the correct solution.

#### S.2.4. Probe Budget and the Squared-Loss Estimand

Let  $A = B^\top \nabla^2 v B$  be symmetric,  $z_k$  independent standard Gaussian vectors, and  $q_k = z_k^\top A z_k$ . Gaussian fourth moments give  $\mathbb{E} q_k = \text{tr } A$  and  $\text{Var}(q_k) = 2 \|A\|_F^2$ . For deterministic  $a$  and  $\bar{q}_K = K^{-1} \sum_k q_k$ ,

$$\mathbb{E} \left( a - \frac{1}{2} \bar{q}_K \right)^2 = \left( a - \frac{1}{2} \text{tr } A \right)^2 + \frac{\|A\|_F^2}{2K}.$$

In contrast  $K^{-1} \sum_k (a - q_k/2)^2$  has excess expectation  $\|A\|_F^2/2$ , independent of  $K$ . For the regression matrix  $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ ,  $\|A\|_F^2 = 15$ ,  $a = 1.25$ , and the exact-trace loss is 1.5625. The population excess is  $7.5/K$ .

An unbiased alternative for  $K \geq 2$  is the off-diagonal statistic

$$U_K = \frac{(\sum_k r_k)^2 - \sum_k r_k^2}{K(K-1)}, \quad r_k = a - q_k/2.$$

Independence implies  $\mathbb{E} U_K = (\mathbb{E} r_k)^2$ . Under an integrable derivative envelope, differentiating its expectation yields the gradient of the exact squared residual. The realized statistic can be negative; unbiasedness neither makes it a nonnegative sample loss nor guarantees optimizer stability. If the action or parameters are selected using the same probes, the fixed-parameter identity is not an adaptive generalization theorem. Independent held-out probes are needed to interpret such selection.

Each row averages 20,000 independent batches. The Monte Carlo standard error pertains to the repeated-batch mean, not to a single quadratic form or its within-batch variance. The numerical test checks the known expectation within six Monte Carlo standard errors and preserves the wrong mean-of-squares statistic as a negative control. It does not assign value or policy errors to this algebraic experiment.

TABLE S.1  
INDEPENDENT-BATCH STOCHASTIC-TRACE LOSSES

| $K$ | Correct mean | Population | MC s.e. | Old mean | Debiased mean |
|-----|--------------|------------|---------|----------|---------------|
| 1   | 8.91810      | 9.06250    | 0.22234 | 8.91810  | –             |
| 4   | 3.47199      | 3.43750    | 0.04758 | 9.17095  | 1.57233       |
| 16  | 2.03146      | 2.03125    | 0.01492 | 9.06287  | 1.56270       |
| 64  | 1.68023      | 1.67969    | 0.00640 | 9.06132  | 1.56307       |

20,000 independent Gaussian batches per row; the exact-trace squared residual is 1.5625. Correct mean averages probes before squaring; old mean averages individual squared residuals. The debiased off-diagonal statistic requires at least two probes. Its standard errors and both variance estimands are retained in the JSON record.

### S.2.5. Executable Computational-Graph Tests

The test suite forms a quadratic critic and a control-dependent diffusion loading, contracts its automatic Hessian with the covariance, and checks both parameter blocks. In the actor update, critic coefficients receive no gradient but the action-dependent covariance does. In the critic update, the policy output is detached while the critic coefficients receive gradients. A finite-transition actor derivative is also checked against central differences in all three controls. Freezing critic parameters does not detach the next-state input: the continuation gradient with respect to the action-induced state is retained.

A separate test compares the resource critic’s explicit gradient with PyTorch automatic differentiation. The convex squared-ReLU feature architecture is a transparent special case of the input-convex approach of [Amos, Xu, and Kolter \(2017\)](#). Exact directional second derivatives or Hessian-vector products, following [Pearlmutter \(1994\)](#), can avoid constructing a dense Hessian when the diffusion has low rank. The complete costs still depend on network size, diffusion multiplication, sample count, probe count, and optimization iterations.

## S.3. THE STOPPED PREFERENCE-FORMATION ECONOMY

### S.3.1. Admissibility and the Boundary Contract

The interior state is  $(u, X) \in (1.2, 2.8) \times (0.5, 2)$ , with the controls and coefficients stated in the main paper. The process is stopped at the first exit or  $T = 1$ , whichever occurs first. It receives  $G(u, X) = -0.02(u - 2)^2 + 0.1 \log X$  at that time. A start on the boundary is settled immediately. This is a finite-horizon contract on an open operating domain; it is not a claim that the unstopped diffusion remains in a rectangle.

For bounded progressively measurable controls, the wealth SDE with coefficients linear in wealth and the additive preference SDE have a well-defined solution up to exit. Before exit, wealth and consumption are bounded away from zero. The flow, terminal payoff, and quadratic adjustment cost are bounded on the stopped domain. Thus the additive objective is finite without introducing a reflected local time or a wealth-transfer process.

The original state-constraint interpretation cannot hold with these coefficients. Preference volatility is 0.05 at both endpoints, and the maximal wealth drift at the lower boundary is  $(0.02 + 0.8 \times 0.06)0.5 - 0.05 = -0.016$ . At zero portfolio volatility it is more negative. Neither fact is repaired by projecting an Euler endpoint back onto the

rectangle. The stopped contract retains all interior coefficients and makes the change in boundary economics explicit.

### S.3.2. A Boundary-Compatible Payoff

The choice of  $G$  has a useful analytic check. Its derivatives are  $G_u = -0.04(u - 2)$ ,  $G_{uu} = -0.04$ ,  $G_X = 0.1/X$ ,  $G_{XX} = -0.1/X^2$ , and  $G_{uX} = 0$ . For every admissible action and every  $k \geq 0$ ,

$$\begin{aligned} \ell^a + \mathcal{L}^a G - \rho G &\leq \max_{u \in [1, 2.8]} \frac{0.8^{1-u}}{1-u} + 0.0064 + 0.0043 - 0.00005 + 0.0032846 \\ &< -0.8162. \end{aligned}$$

The consumption bound follows because utility is increasing in consumption. Maximization over  $u$  at  $c = 0.8$  gives the upper endpoint; it can also be checked by the explicit derivative. The 0.0043 term bounds  $0.1[0.02 + 0.06\pi - c/X]$  by using  $\pi \leq 0.8$ ,  $c/X \geq 0.025$ . The omitted portfolio diffusion term is nonpositive, and so are adjustment costs. The last term bounds  $-\rho G$  uniformly.

Stopped Itô integration therefore implies  $J^p \leq G$  for every admissible policy. To obtain boundary continuity from below, fix for example a constant admissible nonzero portfolio share. Its two-dimensional covariance is uniformly positive definite on the closed rectangle because  $X \geq 0.5$  and  $|\varrho| < 1$ . The boundary is regular for this fixed diffusion, and its expected exit time tends to zero as the start tends to a boundary point. Bounded rewards and continuity of  $G$  imply that this policy's stopped payoff tends to  $G$  there. Since its payoff is bounded above by the optimal value and the latter is bounded above by  $G$ , the value has the declared boundary limit. This argument establishes boundary compatibility; it does not assert global classical differentiability at corners.

### S.3.3. Transition, Covariance, and First Exit

For step  $h$ , form four equiprobable sign pairs  $(z_1, z_2) \in \{-1, 1\}^2$ . Let

$$\begin{aligned} d_u &= \theta h + \sigma_u \sqrt{h} z_1, \\ d_X &= [(r + \pi(\mu - r))X - c]h + \pi X \sigma_X \sqrt{h} [\varrho z_1 + \sqrt{1 - \varrho^2} z_2]. \end{aligned}$$

For an interior state  $s$ ,  $\alpha$  is the first fraction in  $[0, 1]$  at which the line  $s + \alpha d$  meets a face, or one if it stays inside. An exit branch has payoff

$$\frac{1 - e^{-\rho \alpha h}}{\rho} \ell(s, a) + e^{-\rho \alpha h} G(s + \alpha d).$$

A surviving branch instead uses the continuation value at  $s + d$  and the full-step flow annuity. At a boundary start the backup is exactly  $G(s)$ . In particular there is no nonzero continuation from a boundary node. The reference evaluator uses nonnegative bilinear interpolation weights that sum to one on surviving branches.

These signs reproduce conditional first moments and the covariance matrix of the written diffusion. Polynomial tests on  $1, u, X, u^2, X^2, uX$  distinguish covariance terms

from drift-product terms of order  $h$ . At the test state and action the covariance is

$$\begin{pmatrix} 0.0025 & -0.002 \\ -0.002 & 0.0256 \end{pmatrix}.$$

The largest generator discrepancy decreases by a factor of ten when  $h$  decreases from  $10^{-2}$  to  $10^{-3}$  and again to  $10^{-4}$ . Preference controls of  $-0.2$  and  $0.2$  give different mean next states even when both displacements are smaller than a spatial grid cell. No nearest-neighbor map erases the control.

The line-exit rule is an approximation to diffusion killing, not exact Brownian bridge detection. Before-correction overshoots, branch-exit frequencies, occupied exit probabilities, and post-exit payoffs are distinct recorded objects. A reflection-regulator field is inapplicable and is represented by null with that explanation. For the joint state-time refinement, the squared largest mesh width divided by  $h$  decreases as  $0.04, 0.02, 0.01$ ; the separate control refinements are also retained. These observations support consistency diagnostics but do not supply a numerical bound for every  $\kappa_n$  in the main theorem.

#### S.3.4. *Economic Comparative Statics*

For each admissible policy let  $B(p)$  denote cost-free benefit. Define discounted cumulative effort by  $C(p) = \mathbb{E} \int_0^{\tau_p} e^{-\rho t} \theta_t^2 / 2 dt$ . The stopping time is policy dependent but not directly dependent on  $k$  for a fixed policy. Thus  $V(k) = \sup_p [B(p) - kC(p)]$  is decreasing and convex. For optimizers  $p_1, p_2$  at  $k_1 < k_2$ , revealed optimality gives

$$\begin{aligned} B(p_1) - k_1 C(p_1) &\geq B(p_2) - k_1 C(p_2), \\ B(p_2) - k_2 C(p_2) &\geq B(p_1) - k_2 C(p_1). \end{aligned}$$

Adding implies  $(k_2 - k_1)[C(p_2) - C(p_1)] \leq 0$ . Also  $-C(p)$  is a subgradient of  $V$  at any  $k$  for which  $p$  is optimal, and  $V'(k) = -C(p)$  wherever  $V$  is differentiable. For policies with value losses at most  $e_1, e_2$ , the same inequalities with these errors imply

$$C(p_2) - C(p_1) \leq \frac{e_1 + e_2}{k_2 - k_1}.$$

The result applies to the continuous stopped model and separately to each exactly specified positive finite approximation. A finite-grid comparison cannot set  $e_i = 0$  for the continuous problem merely because its discrete maximization was exhaustive.

The reported effort is computed both backward and by forwarding the induced state probabilities. The latter multiplies live branch mass by bilinear weights and accumulates discounted cost through the line-exit fraction. This is an occupation calculation for the interpolation-defined finite Markov chain. It provides an economically interpretable sum rather than an unweighted mean of control magnitudes. A reduction of this aggregate does not imply a pointwise reduction of  $\theta$ , consumption, or portfolio share. Nor does a comparative static identify  $k$  from observations without a specified data model and an injective moment map.

## S.4. PORTFOLIO AND RECURSIVE-UTILITY VERIFICATION

S.4.1. *Stationary Merton Benchmark*

Write  $b = 1 - \gamma$ ,  $V(X) = AX^b/b$ ,  $c = mX$ , and

$$g(m, \pi) = r + (\mu - r)\pi - m + \frac{1}{2}(b - 1)\sigma^2\pi^2.$$

For exponential utility discount  $\rho$ , the fixed-policy stationary equation is

$$A = \frac{m^b}{\rho - bg(m, \pi)}$$

on the positive branch  $\rho - bg > 0$ . Optimization gives  $\pi^* = (\mu - r)/(\gamma\sigma^2)$  and

$$m^* = \frac{\rho - (1 - \gamma)[r + (\mu - r)^2/(2\gamma\sigma^2)]}{\gamma}.$$

At  $(\rho, \gamma, \mu, r, \sigma) = (0.04, 2, 0.08, 0.02, 0.2)$  the targets are  $(m^*, \pi^*) = (0.04125, 0.75)$ . Wealth remains positive under this proportional policy. Its discounted terminal candidate expectation decays at rate  $\rho - bg(m^*, \pi^*) = m^* > 0$ , verifying its own transversality condition. The usual verification comparison is made over policies for which the stopped local martingale and discounted terminal inequalities pass to the limit.

The executed feature solver treats  $q = \log A$ , the consumption ratio, and the portfolio ratio as unknowns. It alternates the fixed-policy scalar evaluation root with an automatic-differentiation actor optimization, without feeding the analytical targets into the updates. Positivity is imposed by  $A = e^q$ . This is a learned homothetic feature model; the ratios alone would not be a neural-function benchmark on an unrestricted state space. The no-short regression at  $\mu = 0.01$  instead has unconstrained share  $-0.125$  and constrained share zero at every positive wealth. It is not a wealth-dependent switching problem.

S.4.2. *Epstein–Zin Driver and the Admissible Comparison Class*

Use  $a = 1 - 1/\psi$ ,  $b = 1 - \gamma$ , with  $a, b \neq 0$ ,  $c > 0$  and  $bV > 0$ . The driver is

$$f(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}.$$

The log limits must be taken separately when a denominator vanishes. For a homothetic candidate  $V = AX^b/b$ , dividing the stationary HJB by  $AX^b$  gives

$$\frac{\rho}{a} \{m^a A^{-a/b} - 1\} + g(m, \pi) = 0.$$

Thus a fixed proportional policy has a unique positive homothetic coefficient whenever  $1 - ag/\rho > 0$ . Its consumption first-order condition is  $\rho m^{a-1} A^{-a/b} = 1$ , and the portfolio first-order condition gives the same Merton share with risk aversion  $\gamma$ . Eliminating  $A$  yields

$$\pi^* = \frac{\mu - r}{\gamma\sigma^2},$$

$$m^* = \psi\rho + (1 - \psi) \left[ r + \frac{(\mu - r)^2}{2\gamma\sigma^2} \right],$$

$$A^* = (\rho(m^*)^{a-1})^{b/a}.$$

For  $(\rho, \gamma, \psi, \mu, r, \sigma) = (0.04, 5, 1.5, 0.08, 0.02, 0.2)$  this gives  $m^* = 0.0455$  and  $\pi^* = 0.3$ . The normalized Hamiltonian is strictly concave in the consumption ratio and portfolio share on the specified positive domain. The selected ratios lie inside the implemented action bounds.

For precision about infinite horizon, the verification class consists of policies for which the policy-wise recursive equation exists on the branch  $bY > 0$ , its localized comparison argument is integrable, and the comparison-weighted terminal remainder vanishes. On compact subsets of  $c > 0, bY > 0$  the driver is locally Lipschitz, but this does not imply a global Lipschitz constant on the entire half-line. One may first stop at compact localization sets, apply the mean-value linearization in the value argument, and then impose uniform integrability and the weighted terminal limit to remove localization. With  $\Gamma_t = \exp(\int_0^t f_V(c_s, \zeta_s) ds)$  for the intermediate value  $\zeta_s$ , the required remainder is of the form  $\mathbb{E}[\Gamma_T |V(X_T) - Y_T^p|] \rightarrow 0$ . The candidate HJB inequality and this comparison yield  $Y_0^p \leq V(X_0)$ , with equality for the maximizing policy.

For the selected homothetic policy itself,  $f_V = -0.1115$ , while the growth exponent of  $\mathbb{E}X_t^b$  is  $bg = 0.066$ . The candidate comparison-weighted value therefore decays at rate  $0.1115 - 0.066 = 0.0455 > 0$ . The numerical record evaluates the fixed-policy equation and target errors, and the analytic calculation verifies this decay. This specifies a genuine stationary recursive benchmark with an explicit admissible verification class; it does not assert that all bounded controls satisfy the terminal condition automatically.

The sign-enforcing architecture is actually executed:  $V = e^q X^b / b$ , hence  $bV = e^q X^b > 0$ . A separate finite-horizon architecture test uses

$$V_\xi(t, X) = \frac{1}{b} \exp\{\log[bG(X)] + (T - t)N_\xi(t, X)\}$$

for  $bG > 0$ . It verifies the terminal identity and sign on test points. It is a domain test, not evidence that an additional finite-horizon recursive problem was trained. These distinctions follow the policy-wise recursive-utility formulation of [Duffie and Epstein \(1992\)](#) rather than replacing recursion by a pointwise evaluation at  $c = 1, V = -1$ .

#### S.4.3. A Structured Stochastic Dynamic Control Check

For  $d \in \{4, 8, 16\}$ , the additional six-period LQR problem uses

$$A = 0.82I + 0.07(J + J^\top), \quad B = 0.2I, \quad Q = I + 0.31\mathbf{1}^\top/d, \quad R = 0.15I,$$

and Gaussian shock loading  $\Sigma = 0.08(I + 0.31\mathbf{1}^\top/d)$ , with terminal matrix  $Q$ . Here  $J$  has ones on its upper adjacent diagonal. The linear neural actor and quadratic feature critic are trained by policy evaluation and improvement. The independent reference uses the finite-horizon Riccati recursion, including the noise contribution to the value constant. A second recursion evaluates the learned fixed policy rather than using its fitted critic as a performance score.

The experiment has cross-state dynamics and non-diagonal stochastic covariance but no binding control constraints. It is retained as a structured dynamic regression test. The main shared-budget resource experiment supplies the separately executed constrained case; neither is relabeled as an adaptive sparse-grid comparison.

TABLE S.2  
STRUCTURED STOCHASTIC LQR REGRESSION

| $d$ | Actor coeff. error | Cost excess max. | Neural sec. | Riccati sec. |
|-----|--------------------|------------------|-------------|--------------|
| 4   | 4.25e-06           | 3.03e-11         | 0.0571      | 0.00037      |
| 8   | 2.25e-06           | 2.14e-11         | 0.0724      | 0.00028      |
| 16  | 4.30e-06           | 3.84e-11         | 0.1330      | 0.00033      |

Six periods, linear actors and quadratic feature critics, independent Riccati reference, and separate frozen-policy evaluation. Cost excess is normalized per dimension in the implementation. This structured unconstrained regression is not the main binding-resource experiment and supplies no generic neural speed advantage.

## S.5. TEMPORAL SELVES AND DYNAMIC COMPETITION

### S.5.1. *Sophisticated Consumption–Savings*

Let one period have length  $\Delta$ ,  $\delta = e^{-\rho\Delta}$ , and terminal payoff  $G(w) = \log w$ . The acting self chooses  $c \in (0, w)$  to maximize  $\Delta \log c + \beta \delta V_{n+1}(w - c)$ , whereas ordinary continuation evaluates the chosen policy as

$$V_n(w) = \Delta \log c_n(w) + \delta V_{n+1}(w - c_n(w)).$$

If  $V_{n+1} = A_{n+1} \log w + B_{n+1}$ , strict concavity gives

$$\begin{aligned} z_n &:= \frac{c_n(w)}{w} = \frac{\Delta}{\Delta + \beta \delta A_{n+1}}, \\ A_n &= \Delta + \delta A_{n+1}, \\ B_n &= \Delta \log z_n + \delta A_{n+1} \log(1 - z_n) + \delta B_{n+1}. \end{aligned}$$

Starting with  $A_N = 1, B_N = 0$  yields an exact sophisticated equilibrium. The coefficient  $A_n$  is independent of  $\beta$ , but the policy and the constant are not. The reference test uses  $\Delta = 1$ , four consumption dates,  $\rho = 0.04$ , and  $\beta \in \{0.7, 1\}$ . At four wealth levels it separately solves the bounded one-shot consumption problem and checks the evaluation identity with the frozen continuation coefficients.

For two consumption dates,  $\delta = 1$ , the correct first-date ratio is  $1/(1 + 2\beta)$ . Reusing the acting self's maximand as its stored continuation would instead yield  $1/(1 + \beta + \beta^2)$ . At  $\beta = 0.7$  the difference is 0.4166667 versus 0.4566210. The  $\beta = 1$  check alone cannot reveal this error. These checks implement the written temporal game rather than an ordinary geometric-discount control problem. They do not identify the finite-state reference as a trained neural solution.

### S.5.2. *A Stochastic Capacity Duopoly*

There are four decision dates. Demand is  $D \in \{1, 1.3\}$  with probability 0.8 of staying at its current value. Each firm's installed capacity  $K_i \in \{0, 1\}$  determines current output  $q_i = K_i/3$ . Current operating profit is  $q_i(D - q_i - q_j)$ . Firm  $i$  chooses a maintenance or renewal intensity  $a_i \in [0, 1]$  at cost  $0.4a_i^2/2$ . Independently across firms conditional on actions, next-period capacity is one with probability  $a_i$  and zero otherwise. The discount factor is 0.95 and terminal value is  $0.05K_i$ . There are eight Markov states and four dates, so complete state enumeration is possible.

For fixed rival intensity, the continuation value is affine in a firm's own intensity. Its best response is therefore the projection onto  $[0, 1]$  of its discounted marginal continuation gain divided by 0.4. The joint map is affine before projection, with slope determined by the cross difference of the four next-capacity continuation values. The program checks the product of the two own best-response slopes, iterates from multiple initial profiles, and compares their limits. This finite example has a contraction bound below one. It does not establish uniqueness in arbitrary dynamic games.

An independent backward best-response recursion freezes the rival's entire state-contingent policy at every subsequent date and allows the tested firm to deviate at all dates. Its difference from the profile value is dynamic exploitability. This is stronger than checking a single current Hamiltonian against an inconsistent continuation. The report records the maximum over all dates, states, and players, with a numerical tolerance of  $10^{-10}$ . A separate automatic-differentiation test confirms that a player's loss does not differentiate the rival's parameters.

The static special case  $q_i(1 - q_i - q_j)$  remains a useful negative control. Symmetric Nash output is  $1/3$ , whereas joint-profit output is  $1/4$ . At the latter profile, changing one firm's quantity to its own best response is profitable. A symmetric or jointly optimal actor graph is therefore not a substitute for a Nash calculation. The dynamic capacity model is a finite-state equilibrium computation; no competitive limit or general neural-game convergence is inferred from it.

### S.5.3. Market-Size Normalization and a Conditional Small-Player Bound

To retain the economically relevant many-firm extension without conflating different economies, consider per-firm normalized output  $y_i$  and actual output  $q_i = y_i/N$ , with inverse demand  $P = a - b\bar{y}$ ,  $\bar{y} = N^{-1} \sum_i y_i$ . Multiplying each firm's monetary payoff by  $N$  gives normalized flow  $y_i P - C(y_i)$  under consistently scaled costs. Its direct strategic price derivative is  $-by_i/N$ . Investment, depreciation, and state transitions must be scaled and retained as well; this current-flow derivative by itself does not establish convergence of dynamic equilibria.

A precise conditional implication is available. Suppose that, against a fixed rival profile, the finite- $N$  player's transition operators and a price-taking comparison operator are monotone and  $\gamma_n$ -Lipschitz, have the same terminal payoff, and differ uniformly by at most  $\eta_n/N$  on all policies and continuation values used in the comparison. Then a policy optimal for the price-taking problem loses at most

$$\frac{2}{N} \sum_{j=n}^{N-1} \Gamma_{n,j} \eta_j$$

relative to the finite- $N$  best response. To prove it, backward iteration bounds the difference of the two optimal values by  $N^{-1} \sum_j \Gamma_{n,j} \eta_j$ , and bounds the two evaluations of that fixed policy by the same amount. Their sum gives the result. Uniform Lipschitz control of aggregate-state effects is one route to the operator hypothesis, but it must be checked in a specified model. This proposition concerns a unilateral deviation bound, not selection or convergence of a sequence of equilibria. It is theoretical extension material and is not recorded as an executed large- $N$  experiment.

### S.6. REPLICATION AND ADDITIONAL RESULTS

This section retains the foundational neural experiments and their full checkpoint provenance. The fifth-revision additions and independent rechecks are described in S.7–S.8.

#### S.6.1. *Source and Evidence Identities*

The authoritative R4 source record consists of `ECTA_R4.tex`, `SUPP_R4.tex`, their explicitly included sections, and `replication/r4/`. The base review is pinned at commit `79a7d84`. The replication manifest records the reachable source commit, full SHA-256 digests of the source files, package versions, and output identities. Source files are committed before their authoritative execution. A later result commit can add generated outputs without changing that source anchor.

Execution status, evidence class, and numerical-tolerance status are separate fields. A completed program is not automatically a solved control problem. The coupled resource runs have a specified held-out cost-loss target and independent reference gaps. The NDU comparison reports discrepancies and does not assign a passed uniform PDE certificate. Algebraic checks, domain tests, finite-state equilibria, and structured feature solvers each have their own evidence class. An unavailable continuous-time error bound or reflection regulator is null with an explanation, not a numerical zero.

Scientific digests exclude elapsed time and hardware measurements. File digests still identify complete files, including timing records. Floating-point algorithms, package versions, and BLAS implementations can change last digits across platforms. The source and saved weights permit direct policy replay independently of retraining. Timings include neural fitting and full evaluation for the stated initial-state batch; they do not imply a universal hardware comparison or unmeasured amortization advantage.

#### S.6.2. *Architectures and Optimization*

For the stopped preference model the safeguarded neural run uses separate two-hidden-layer tanh actors of width 32 and critics of width 64 at each of eight decision dates. The actor has three bounded output coordinates. The critic writes the value as  $G$  plus remaining time times a product vanishing on all spatial faces times its network output. The training sample contains 2,048 interior states. L-BFGS caps are 160 actor and 1,000 critic iterations with strong-Wolfe line search; actual function evaluations and iteration counts are recorded when produced by the optimizer. The implementation never trains on the independent grid reference’s values or policies.

The unsafeguarded pilot uses width-32 critics and 1,024 training states with a smaller critic budget. It is retained as a development baseline, not used as a matched causal ablation. The resource experiment supplies a matched ablation: the same fitted actor weights are evaluated with and without pointwise improvement at deployment. It holds training, initial states, shocks, and model primitives fixed.

For resources the convex critic is

$$q(x) + \ell^\top(x - \mathbf{1}) + c_0 + \sum_{j=1}^{6d+102} \alpha_j [\max\{w_j^\top(x - \mathbf{1}) + b_j, 0\}]^2, \quad \alpha_j \geq 0.$$

Directions consist of signed coordinate axes, signed aggregate directions, and 32 seeded Gaussian directions, each at offsets  $-0.5, 0, 0.5$ . The coefficients are fitted by bounded

TABLE S.3  
ALL RESOURCE RUNS AND THE MATCHED DEPLOYMENT ABLATION

| $d$ | Seed | NBO upper loss | Proposal only | Reference gap | Binding at 0 |
|-----|------|----------------|---------------|---------------|--------------|
| 4   | 101  | 6.45e-04       | 1.19e-02      | 8.32e-08      | 0.812        |
| 4   | 202  | 4.88e-04       | 8.14e-03      | 7.40e-08      | 0.938        |
| 4   | 303  | 5.23e-04       | 6.31e-03      | 7.02e-08      | 0.938        |
| 8   | 101  | 4.29e-04       | 9.54e-02      | 9.79e-08      | 0.938        |
| 8   | 202  | 4.61e-04       | 3.28e-02      | 9.97e-08      | 1.000        |
| 8   | 303  | 2.03e-04       | 3.89e-02      | 9.75e-08      | 1.000        |
| 16  | 101  | 5.41e-04       | 6.73e-02      | 9.54e-08      | 1.000        |
| 16  | 202  | 5.67e-04       | 6.81e-02      | 9.63e-08      | 1.000        |
| 16  | 303  | 5.24e-04       | 6.65e-02      | 6.02e-08      | 1.000        |

Each row uses the stored 32 initial states and every one of the 64 terminal shock branches. The reference gap is the complete scenario-tree linear minimization gap at the tighter audit tolerance. Binding is the fraction of initial actions exhausting the common budget. All declared dimension–seed combinations are included.

least squares with a  $10^{-5}$  ridge term. The actor has one width-32 tanh layer and a slack-softmax output enforcing the common budget. Its 160-iteration L-BFGS fit distills feasible greedy actions. The convex pointwise improvement uses an explicit gradient-Lipschitz upper bound and checks its linear minimization gap rather than equating a stationary point with a maximum. All three seeds, 101, 202, and 303, are included at each dimension.

The independent scenario tree contains one root action, four next-date actions, and sixteen last-date actions for each tested initial state. Its objective sums all state and control costs and the 64 terminal costs with their probabilities. An independent backward adjoint computes the gradient. Reference optimization stops at a common-accuracy gap of  $10^{-3}$  for timing and is separately continued to  $10^{-7}$  for policy-cost auditing. Nonanticipativity is enforced by one action per information node, not one freely chosen action per complete shock path.

### S.6.3. *Reproduction Order*

The repository README gives executable commands for the following sequence: generator, temporal, trace, utility, and game tests; stochastic grid refinements and re-solved adjustment-cost panels; the retained unsafeguarded neural pilot; all three safeguarded NDU runs; nine resource runs; independent frozen-policy and occupation analysis; computational-graph and checkpoint-replay tests; table generation; source/output manifest generation; and LaTeX builds. The table generator reads the deposited JSON records, so manually entered historical arrays cannot silently become current numerical evidence.

The source snapshot also retains the previous manuscripts, supplements, reports, reviewer diagnostics, and arrays unchanged. The preservation map locates their substantive topics in the new manuscript or this supplement. Retention is not endorsement of earlier incorrect claims. The development log records the standard-library module-name collision discovered during implementation, the corrected execution, and the budget difference of the pilot. Failed commands are not relabeled as successful scientific experiments.

### S.6.4. *What Each Numerical Error Means*

For NDU, a reference value difference compares two specified finite approximations or evaluates a neural policy against a specified finite reference. The actor discrepancy reports consumption, preference adjustment, and portfolio separately because they have different units. A feasible Bellman gain is measured in continuation-value units and uses an independently evaluated frozen continuation. It is a lower bound on the maximum profitable one-step deviation over a richer action set, not an upper bound on all deviations.

For resources, the full stochastic tree gives the evaluated policy cost. The reference tangent gap bounds its unknown optimum on that same tree, resulting in the interval in S.1. The held-out initial states are explicitly stored, so the bound is not extrapolated to all states. The Merton and Epstein–Zin policy errors compare learned homothetic coefficients with analytical solutions, while LQR errors compare structured learned policies with an independent Riccati solution. Temporal and game exploitability use separate best-response calculations against the actual continuation being evaluated. Trace standard errors concern a repeated-batch scalar loss and are not policy errors.

## S.7. STOPPED CONTRACTS AND VALUE-ENCLOSED POLICY REUSE

### S.7.1. *Discounted Duration and Terminal Exposure*

For any realized stopping time  $\tau \leq T$ , direct integration gives

$$\rho \int_t^\tau e^{-\rho(v-t)} dv + e^{-\rho(\tau-t)} = 1.$$

This equality holds before conditioning and before optimizing. In the positive finite model, each branch satisfies the same identity with its actual live duration  $\alpha h$ . Thus the independently evaluated exposure recursion is

$$Z_n^p(s) = z_n(s, p_n(s)) + K_n^{p_n} Z_{n+1}^p(s), \quad Z_N^p = 1,$$

where  $z_n$  is the discounted probability weight of settlement during the current branch and  $K_n^p$  contains only live discounted interpolation weights. Duration satisfies  $A_n^p = a_n^p + K_n^p A_{n+1}^p$ ,  $A_N^p = 0$ . The row identity  $\rho a_n^p + z_n^p + K_n^p \mathbf{1} = 1$  proves  $\rho A_n^p + Z_n^p = 1$  by induction. This includes immediate settlement for an initial boundary node.

The program does not define  $Z$  as  $1 - \rho A$  and then test the resulting tautology. It evaluates  $Z$  with its own killed-payment recursion and checks the identity for randomly selected contracts and stored-policy switches. It then compares the original four-feature payoff  $B + mA - kC + \ell Z$  with the reduced payoff  $\ell + B + (m - \rho\ell)A - kC$ . The largest discrepancy and the source hash are recorded. For state paths generated by an optimal-policy selection, all changes  $(m, \ell) \mapsto (m + \rho h, \ell + h)$  preserve the set of feasible optimal laws; this is the precise sense in which choices alone do not identify the two levels separately.

The monotonicity result concerns a common transition and feasible-policy class. Changing the shock correlation, the control bounds, or the stopping mechanism does not satisfy that condition. Nor does a different parameter-specific action menu necessarily preserve convexity of the optimized finite value in the original parameters. We therefore freeze the complete union menu over the contract interval before building the library.

### S.7.2. Certified Anchors and Approximate Policy Features

Let a feasible anchor policy  $p_i$  have certified loss  $c_n^i(s)$  at its anchor  $q_i$ . Exact evaluation gives the valid upper bound  $\bar{V}_n^i = J_n^i(q_i) + c_n^i$ . If evaluation instead supplies an interval  $[J_n^i, \bar{J}_n^i]$ , the valid anchor upper bound is  $\bar{J}_n^i(q_i) + c_n^i$ . The empirical maximum of sampled actor residuals cannot be substituted for either ingredient.

For clarity, an error version of switching can be stated without conflating approximate policy features with exact evaluation. Suppose  $\|\hat{J}_n^i - J_n^i\|_\infty \leq e_n$  uniformly over stored policies at the query, and choose the policy index maximizing  $\hat{J}_n^i$ . Let  $L_n = \max_i J_n^i$  and  $D_n = \|(L_n - J_n^{\text{sw}})_+\|_\infty$ . If continuation kernels have row sums at most  $\gamma_n$ , then

$$D_n \leq 2e_n + \gamma_n D_{n+1}, \quad D_n \leq 2 \sum_{j=n}^{N-1} \Gamma_{n,j} e_j \quad (\text{S.1})$$

with common exact terminal evaluation. To prove the first inequality, the selected policy has exact value at least  $L_n - 2e_n$ . The next switching value is at least the selected continuation value minus  $D_{n+1}$ ; positivity gives the result. Hence

$$V_n^* - J_n^{\text{sw}} \leq U_n - L_n + 2 \sum_{j=n}^{N-1} \Gamma_{n,j} e_j.$$

The exact-feature theorem in the main paper is the  $e_n = 0$  case. The executed finite recursion uses this case up to recorded floating-point errors; it does not assign zero error to the neural critic. This distinction allows neural approximation to provide proposals while the certificate uses separately evaluated policies.

### S.7.3. A Finite Certificate for an Entire Interval

For adjacent anchors  $a < b$ , two feasible endpoint-policy lines are  $b_1 + da_1$  and  $b_2 + da_2$ , independently at every date and state. With an affine upper chord  $u_0 + du_1$ , define

$$g(d) = u_0 + du_1 - \max\{b_1 + da_1, b_2 + da_2\}.$$

On either side of the intersection of the two lines,  $g$  is affine. Its maximum on  $[a, b]$  must therefore occur at  $a$ , at  $b$ , or at their intersection if present. Parallel lines require only the endpoints. Taking the maximum over the finite set of dates and states gives a uniform bound without discretizing the contract interval. Enlarging the policy library can only improve the maximum of feasible policy values; using just the endpoint pair to certify an interval is conservative and reduces certification cost.

The adaptive construction begins with contracts zero and one. At every iteration it computes the complete bound, selects a worst interval, and solves a new anchor at a parameter attaining that interval's largest gap. Each new policy is independently evaluated. The complete anchor sequence, worst gaps, and parameter locations are saved, including the stopping criterion. The implementation has a maximum-anchor safety limit that raises an error rather than publishing an uncertified library when the tolerance is not reached.

All computations use double precision. The pairwise-intersection calculation treats duration-slope differences below  $10^{-14}$  as parallel. The omitted variation of a difference of such lines on the unit contract interval is at most  $10^{-14}$ ; the verification checks

use a larger  $10^{-9}$  slack. The published  $10^{-3}$  threshold has substantial slack relative to both quantities. This is a numerical certificate based on explicit finite inequalities, not an interval-arithmetic proof of every floating-point operation. Independent policy recursions, kernel comparisons, and direct query solves test the implementation of the inequalities.

#### S.7.4. *Whole-Interval Action Separation*

At a fixed date and state, choose one feasible first action  $a$  and one library continuation  $p_i$ . Both its reward and its continuation value are affine in the contract. For each competing first action  $b$ , the upper-chord continuation gives another affine function. If their difference is positive at both interval endpoints, it is positive at every convex combination of the endpoints. Taking a supremum over all competing actions then proves the main action-separation proposition.

The implementation maximizes the minimum of the two endpoint margins over candidate pairs  $(a, p_i)$ . This ordering of maximization and minimization is essential. Taking the minimum of two independently maximized endpoint margins would permit a different favorable witness at each endpoint and would not establish a single affine lower bound on the interval. Saved output includes the selected library index, the first action, both sign margins, and the reported status of each interval. The displayed negative-region endpoint is rounded downward and the positive-region endpoint upward; neither rounded interval extends beyond its certificate.

#### S.7.5. *Relation to Existing Policy-Transfer Results*

The feature decomposition and the policy-improvement logic build on [Barreto et al. \(2017\)](#), [Chang \(2021\)](#). The anchor upper bound is the convex-combination case of [Nemecsek and Parr \(2021\)](#), whose policy-cache method already addresses whether an existing policy is sufficiently accurate for a new reward function. Our construction uses that bound on the economically reduced stopped-contract parameter, checks every interval breakpoint and every finite state-date node, and applies a separate interval-wide action-separation inequality. The independently verified finite-economy results and the stopping-contract interpretation are the substantive objects reported here. We do not claim that the underlying supremum-of-affine-functions argument, policy caching, or actor-critic separation was first introduced in this paper.

### S.8. ADDITIONAL COUNTERFACTUAL AND EQUILIBRIUM COMPUTATIONS

#### S.8.1. *Joint Refinement and Contract Sensitivity*

The complete refinement panel is reported in Table [S.4](#). Unlike refining time on a fixed interpolation grid, these calculations change state spacing, time spacing, and the action mesh together. They retain the consumption and preference controls as active optimization coordinates. The coarse, intermediate, and fine initial values at  $d = 0$  are different; the portfolio at the midpoint is also sensitive to the first refinement. The endpoint contract comparison is more stable than the estimated switching location. All entries remain evaluations of declared finite economies rather than upper and lower bounds on the stopped diffusion.

The correlation panel in the machine-readable output independently resolves the intermediate economy at correlations  $-0.25$ ,  $0$ , and  $0.25$ . For each, the initial share

TABLE S.4  
JOINT TIME, STATE, AND ACTION REFINEMENTS

| State grid     | Dates | Action grid            | $d$ | Value     | Duration | $\pi_0$ |
|----------------|-------|------------------------|-----|-----------|----------|---------|
| $17 \times 25$ | 4     | $5 \times 5 \times 7$  | 0.0 | -1.140548 | 0.884705 | -0.50   |
| $17 \times 25$ | 4     | $5 \times 5 \times 7$  | 0.5 | -0.698148 | 0.885220 | -0.50   |
| $17 \times 25$ | 4     | $5 \times 5 \times 7$  | 1.0 | -0.253215 | 0.891936 | 0.80    |
| $33 \times 49$ | 8     | $7 \times 7 \times 11$ | 0.0 | -1.136633 | 0.887830 | -0.50   |
| $33 \times 49$ | 8     | $7 \times 7 \times 11$ | 0.5 | -0.692395 | 0.890237 | 0.80    |
| $33 \times 49$ | 8     | $7 \times 7 \times 11$ | 1.0 | -0.246813 | 0.894453 | 0.80    |
| $49 \times 73$ | 16    | $9 \times 9 \times 13$ | 0.0 | -1.133946 | 0.888358 | -0.50   |
| $49 \times 73$ | 16    | $9 \times 9 \times 13$ | 0.5 | -0.689493 | 0.890188 | 0.80    |
| $49 \times 73$ | 16    | $9 \times 9 \times 13$ | 1.0 | -0.243752 | 0.894848 | 0.80    |

All controls are reoptimized at every resolution. These refinements use the displayed meshes without the extra frozen neural proposals. The endpoint reversal survives, whereas the midpoint changes sign between the first and second resolution. This table measures sensitivity of finite economies, not a certified bound for the continuous stopped diffusion.

TABLE S.5  
ADJUSTMENT QUANTITY AND COEFFICIENT-WEIGHTED EXPENDITURE

| $k$ | $C^{p_k}$ | $kC^{p_k}$ | Value     |
|-----|-----------|------------|-----------|
| 0.5 | 0.014614  | 0.007307   | -1.117524 |
| 2.0 | 0.009888  | 0.019777   | -1.135835 |
| 8.0 | 0.001432  | 0.011455   | -1.160291 |

The complete finite problem is solved separately at each coefficient on a  $25 \times 37$  grid with eight dates and  $9 \times 9 \times 13$  actions. The unweighted quantity is decreasing; the coefficient-weighted charge is not. Stopping is included in both quantities.

changes from  $-0.5$  at  $d = 0$  to  $0.8$  at  $d = 1$ . We do not transfer the feature bank across these changes in transition law. Duration, values, and all controls are stored for each separate solve. The normalization comparison, by contrast, changes only the reward coefficients and retains the same transition and policy class.

Table S.5 distinguishes discounted adjustment effort from its coefficient-weighted charge. The same decomposition is evaluated at the original cost-panel grid and with the original complete control vector. The theorem applies to the first quantity; the middle coefficient produces the largest weighted charge in this panel.

### S.8.2. Matched Actor-Free and Quadratic Comparisons

The nine actor-free comparisons reload the deposited nonlinear critic weights and the same initial states. Both actor-start and zero-start algorithms use the same projected convex improvement method, gradient, Lipschitz step bound, and stopping tolerance. The full policy is then evaluated through all 64 terminal shock paths. Saved output separates actor initialization, optimization time, iterations at each date, complete life-time cost, and the reference gap. The original proposal-only ablation remains in S.6; removing the safeguard and removing the actor are different experiments.

To isolate training costs, critic-only backward fits use the same training-state draws and feature construction as the original nonlinear critic. Actor distillation is timed

TABLE S.6  
ISOLATED ACTOR COSTS AND INITIALIZATION SAVINGS

| $d$ | Seed | Actor train | Actor query | Zero query | Projected break-even |
|-----|------|-------------|-------------|------------|----------------------|
| 4   | 101  | 1.483       | 0.0414      | 0.0424     | 48512                |
| 4   | 202  | 0.741       | 0.0432      | 0.0451     | 12544                |
| 4   | 303  | 0.742       | 0.0436      | 0.0429     | None                 |
| 8   | 101  | 0.808       | 0.0607      | 0.0626     | 13568                |
| 8   | 202  | 0.797       | 0.0637      | 0.0713     | 3392                 |
| 8   | 303  | 0.794       | 0.0489      | 0.0512     | 11488                |
| 16  | 101  | 0.878       | 0.1004      | 0.1035     | 8864                 |
| 16  | 202  | 0.879       | 0.1174      | 0.1326     | 1856                 |
| 16  | 303  | 0.875       | 0.1051      | 0.1079     | 9888                 |

Seconds per 32-state complete-tree query batch. Actor training is the distillation fit alone; common target construction and critic training are not charged twice. The last column is  $32 \lceil T_{\text{actor}} / (t_{\text{zero}} - t_{\text{actor}}) \rceil$  when the measured saving is positive. It is a linear fixed-batch projection, not an observed crossover. “None” means that actor initialization was not faster in that timing comparison. Optimizer iteration counts and separated initialization times are saved in the machine-readable results.

separately against the same improved-action targets. Table S.6 reports the complete nine-case timing panel. A positive measured online saving is converted into a clearly labeled fixed-batch break-even projection. The separately executed fresh-state workloads in the main paper, not this projection, support the observed amortization comparison.

The quadratic comparator minimizes squared fitting error over

$$q(x) + b + \ell^T z + z^T H z, \quad z = x - \mathbf{1}, \quad H \succeq 0.$$

The regression is convex in  $(b, \ell, H)$ . Its gradient is computed from the full quadratic design matrix. Projected accelerated gradient leaves  $(b, \ell)$  unrestricted and projects the symmetric matrix onto the positive-semidefinite cone by clipping negative eigenvalues. A small ridge term stabilizes the design. The reported projected-gradient norm is below the declared  $10^{-8}$  stopping threshold in the executed fits. At deployment the analytic gradient is  $\nabla q(x) + \ell + 2Hz$ , and its Hessian bound is the largest eigenvalue of  $\nabla^2 q + 2H$ . The common investment projection and full scenario-tree cost calculation are unchanged. This comparison tests a genuinely coupled convex quadratic, rather than a separable or unoptimized baseline.

The fresh-state workload charges only for the computations used by each competing method. The reusable policy’s cost is critic training plus zero-start online improvement and full policy evaluation. The specialized solver’s cost is a complete nonanticipative reference optimization at tolerance  $10^{-3}$ . A reference solved more tightly to gap below  $10^{-7}$  supplies the accuracy upper bounds for both; it is not charged asymmetrically as part of one solver’s online cost. Hardware, dependencies, seeds, and individual times accompany the outputs. A three-period, rank-two-shock experiment cannot establish dimension-independent performance, even when its measured query reuse is useful.

### S.8.3. Persistent Capacity and All-Date Dynamic Deviations

Let  $b_i = 1 - sK_i$ , so that next-period capacity probability is  $sK_i + b_i a_i$ . Write  $W_i(k'_1, k'_2)$  for player  $i$ ’s continuation value after averaging over next demand. For player one the

best response is

$$a_1 = \text{clip} \left( \frac{\delta b_1}{\kappa} \{W_1(1,0) - W_1(0,0) + (sK_2 + b_2a_2)\Delta W_1\}, 0, 1 \right), \quad (\text{S.2})$$

where  $\Delta W_1 = W_1(1,1) - W_1(1,0) - W_1(0,1) + W_1(0,0)$ . The corresponding formula holds for player two. The action criterion is strictly concave because the continuation is affine in the player’s own probability and the action cost has curvature  $\kappa > 0$ . Enumerating whether each clipped best response is at zero, interior, or one gives nine active-set combinations. Feasible candidates are checked against both original best-response equations.

The product of the two affine slopes bounds the Lipschitz constant of the composed best response; clipping is nonexpansive. Its recorded maximum below one therefore provides a unique one-period fixed point for every continuation game. Backward induction supplies a finite-horizon Markov-perfect profile. This argument does not assume current-capacity independence and does not rely on convergence from a single initialization.

For independent dynamic verification, let  $J_{i,n}$  be the value of the frozen profile and define the full unilateral best-response value

$$B_{i,n}(s) = \max_{a_i} \{r_i(s, a_i, p_{-i,n}(s)) + \delta \mathbb{E}[B_{i,n+1}(s') \mid s, a_i, p_{-i,n}(s)]\}.$$

At *every* backward date, the code stores  $B_{i,n}(s) - J_{i,n}(s)$  for both players and all eight states. It reports the largest positive gain, the signed maximum and minimum, and the attaining date, player, capacities, and demand. The historical code had placed the comparison after the date loop, which verified only date zero. The corrected scope is executed for both the renewal regression and the persistent economy.

As a negative control, the final-period investment of player one at capacities (1,1) and low demand is replaced by one. The independent routine recomputes the frozen-profile value and all best responses. It detects a gain exceeding 0.1, at a recorded late-date node. This test would not be passed by a routine that merely reports convergence of the equilibrium solver. The reset case  $s = 0$ , four dates, reproduces its early closed-form actions and terminal investment to numerical precision. Both examples remain exact finite-state references, not neural-equilibrium training results.

#### S.8.4. Executable Files and Preservation

The fifth-revision package is in `replication/r5`. It imports the unchanged fourth-revision transition, resource primitives, and saved neural checkpoints. The contract computations are in `contracts.py` and `economic_robustness.py`. The remaining experiments are in `resource_ablation.py` and `persistent_game.py`. The table generator reads executed JSON; the validation script replays policy features, normalization identities, kernel comparisons, and certificate conditions. The build records input and output hashes and preserves the exact review-base commit.

Earlier papers, theorem proofs, raw negative controls, referee reports, and neural checkpoints remain in their original paths. The new manuscript carries the original mathematical and economic extensions forward and adds the contract-reuse and persistent-capacity material. The accompanying response identifies where each issue in the fourth-revision report is treated. Build completion and numerical verification do not assert journal submission or editorial approval.

## S.9. TRANSITION-LAW CERTIFICATES AND MECHANISM REPLICATION

## S.9.1. Coefficient Construction and Interval Bounds

For a fixed policy, pre-draw the  $h$  remaining iid regime labels. Conditional on their count  $j$ , labels are exchangeable, the first label is 1 with probability  $j/h$ , and the remaining count is  $j - 1$  or  $j$ . Applying the same count-blind policy to these conditional paths gives the main coefficient recursion. Averaging over the binomial count distribution proves the polynomial representation. This is an evaluation identity, not a policy allowed to condition its action on the count.

For the separate upper calculation the controller is allowed to observe the remaining count, and past labels are revealed after their transitions. Every original policy can ignore this extra information. Backward maximization of the weighted common-action criterion therefore gives an upper coefficient vector. Averaging over counts preserves the upper inequality. Optimizing the regime-0 and regime-1 actions separately *before* taking their weighted mean would instead disclose the current label and define a different relaxation. The implementation takes the maximum after the mean.

If  $h \geq 2$ , the degree- $h$  Bernstein coefficients of the corrected upper value on a local interval are

$$U_{n,j} = \left(1 - \frac{j}{h}\right) v_n^a + \frac{j}{h} v_n^b + \frac{j(h-j)}{h(h-1)} M_n, \quad j = 0, \dots, h.$$

At  $h = 1$  the correction is zero because the common terminal payoff has zero endpoint difference; at  $h = 0$  the only coefficient is  $g$ . Degree elevation thus represents the quadratic upper function exactly. De Casteljau subdivision restricts both upper and lower coefficients to the same closed cell before taking their coefficient differences. The bound is conservative because  $\sup_t \min_p f_p(t) \leq \min_p \sup_t f_p(t)$ . The minimum is taken separately for each state and date, not before the statewise maximum.

The stopping condition uses the maximum of these cell bounds, not the largest loss observed at replay parameters. The replay additionally checks the whole value vector, dominance by the upper function, and improvement from switching relative to the best stored fixed policy. The script stores each anchor policy, its independently evaluated polynomial coefficients, each interval's upper coefficients, cell bounds, and validation locations. The count-information and corrected-chord outputs remain separate so that a loose information bound cannot be substituted for the stopping certificate.

## S.9.2. A Reusable QP with the Original Information Structure

At a tree node, write its state as  $x_h = M_h x + S_h z + b_h$ . Let  $Q = I/d + 0.211^\top/d^2$  and let  $p_h$  be its probability. Expansion of all stage and terminal costs gives

$$H = 2 \sum_h p_h S_h^\top Q S_h + 2 \text{diag}_h(0.1 p_h I/d), \quad F = 2 \sum_h p_h S_h^\top Q M_h, \quad f_0 = 2 \sum_h p_h S_h^\top Q (b_h - \mathbf{1}).$$

The control diagonal includes decision nodes only. The sums over state costs include terminal nodes. Let  $D$  repeat each decision node's probability across its  $d$  controls. A Lipschitz constant for the transformed gradient is the largest eigenvalue of  $D^{-1/2} H D^{-1/2}$ . Each accelerated step is a nodewise simplex projection of  $z - D^{-1} \nabla C_x(z)/L$ . Every ten iterations we check the complete convex tangent gap; the final gap is independently recalculated by the original tree-gradient routine. The reported query time includes that

evaluation. The structural setup and the learned training times are both included in their respective totals; the tighter deposited reference enters only the accuracy comparison.

The assembly is a new source implementation of the structural comparison described in the fifth-round report, not an invention of parametric quadratic programming. All cold-start workloads use the same saved initial states. No array of known query solutions initializes a new query. Complete costs, gaps, timing samples, feasibility errors, and gradient comparisons are retained in the execution record.

### S.9.3. Risk-Class Evaluation and Scope

For a given  $(d, k)$ , first solve the original finite problem backward. At the initial state, optimize the current action separately over positive and nonpositive portfolio positions using the same optimal continuation. Evaluate each resulting complete policy independently to obtain  $(B, A, C)$ . This computes first-risk-class values, not policies constrained to keep the same sign forever. Repeat the entire backward solve with  $\theta = 0$  at every state and date for the no-deliberate-adjustment model. Preference volatility remains present. The two risk-specific option values are nonnegative by set inclusion, and their difference is checked against the independently formed difference of risk advantages.

Bisection starts only when the risk advantage has opposite endpoint signs. The output records both signs, the bracket, the active class features, and the ratio of their effort and duration differences. It does not assign a global uniqueness flag. A duration-gap lower bound valid throughout an interval would justify the main uniqueness result; the present numerical brackets do not assert that additional hypothesis. The portfolio-sign exclusions in the inherited reward experiment therefore retain their original independent status.

### S.9.4. Evidence and Reproduction

The current source record is `ECTA_R6.tex`, `SUPP_R6.tex`, the sixth-revision section directory, and `replication/r6/`. Sections S.1–S.8 preserve earlier substantive proofs, specifications, and evidence with their historical identities. The new manifest pins the reachable source commit and hashes the executable sources, immutable input checkpoints, generated records, and tables. The public execution log distinguishes source preparation, experiments, validation, and document build.

The support comparison uses `bank_comparison.py`. Transition calculations use `transport.py` and `kernel_certificate.py`. The remaining programs are `mechanism.py`, `structural_qp.py`, and the independent small-model checks in `unit_tests.py`. Numerical libraries are single-threaded for timing. Two developmental transition-certificate runs were stopped to eliminate repeated interval calculations and then to replace array multiplication by an equivalent CSR multiplication. The final runs use the cached intervals and verified CSR rows. This is an execution optimization, not a change in the state, action, transition, or welfare target. The all-state scalar-OLS comparison retains the inherited common oracle implementation for both objectives.

The revised source does not reclassify the historical neural fits, exact grid repairs, or finite games. In particular, removal of neural proposals still meets the original reward-bank welfare target, and the reusable QP still supplies the relevant stronger resource comparator. A completed build or a passing finite-model validation does not establish journal acceptance, broad neural superiority, or a diffusion approximation theorem.

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